

MACHINE LEARNING IN FINANCE

Group examination paper with presentation (all given the same grades) - 30% of the final mark for this course as indicated in the fact sheet of the course.

Project description and guidelines

This assignment is about creating a financial machine learning research project. You need to collect a financial dataset from the broader area of finance **yourselves**. Your dataset should have a good number of predictors (10 or more) and it would be good to have at least few hundreds of observations. Make sure that the financial dataset that you use is **not a time series** one (strictly).

Some guidelines regarding the data search follow. With your HSG Library account, you can access Bloomberg ([Bloomberg | Library University of St.Gallen \(HSG\) \(unisg.ch\)](#)), Eikon ([LSEG Eikon | Library University of St.Gallen \(HSG\) \(unisg.ch\)](#)), and for those having interest in sustainability there are plenty of databases that you could look into via the following link [ESG Databases\[1\] \(unisg.ch\)](#). Please contact a HSG librarian for more information. Moreover, some potential open access data sources you could investigate are (but not limited to) Yahoo Finance, Google Finance, World Bank Open Data, FRED (Federal Reserve Economic Data), European Central Bank Statistical Data Warehouse, IMF Data Portal etc.

You need to **submit your financial dataset in excel format**, the **R code** that you have used, a **pdf file of your research project** (around 10 pages, 1.5 spaced, and 11-pt font), and your **presentation slides** in pdf format.

The **results** from your coding file **should be replicable** (so that anyone having access to your code and data be able to reproduce your results) and match those included in your research project pdf file and presentation slides.

The pdf file of your research project should also contain all references that you have used including any coding or other aids such as ChatGPT. Please search university regulations about this.

You will submit the documents **via email through study net canvas** to despoina.makariou@unisg.ch. In your email, please indicate your group name as appears in Canvas where you have set up your groups (e.g. Group 1, 2, 3 etc and the names of the group participants). The **submission deadline is 14.10.2026 at 17:00 and refers to all files mentioned earlier**. The **presentations will happen on 07.10.2026** during our normal course hours. **Each group presentation should last 15 minutes incl. Q&A**. The **presentations date match those indicated in our first lecture namely Basics of Machine Learning with examples in financial applications**.

Tip: Before finding your financial dataset, it will be of great help to **read the research paper [To Explain or to Predict? \(berkeley.edu\)](#)** as this will help you with **designing your research project (incl. data and variables choice)** and it can provide you with **guidance on how to reply many of the project questions here**.

Questions

Task 1 – Mark: 20%

- Write a short introduction about your financial topic. Please include a thorough academic literature review (ideally based on research published on top finance journals and related outlets) on your chosen financial topic. As you review the literature, clearly indicate whether your study addresses any gaps not yet explored in the literature.
- Define your study goal using predictive modelling terminology and describe its importance from a financial viewpoint. Be clear about whether you perform a classification or regression task.
- Describe your study design and financial data collection process.
- Present and discuss your data (summary statistics) using R.
- Prepare your data for modeling using R (if data pre-processing is needed).
- Perform some exploratory data analysis using R.
- Choose your input and output variables and discuss why this choice is appropriate from a predictive modelling perspective.

Task 2 – Mark: 40%

- Describe the machine learning methodology that you are using. This will be a random forest.
- Train a random forest using the randomForest package in R using default hyperparameters.
- Find your optimal hyperparameters. Specifically, perform tuning for the hyperparameters: number of trees and number of variables to be considered at each split using the caret package in R making use of the resampling techniques namely 10-fold cross validation and leave one out cross validation.
- Choose your final random forest model and make predictions.
- Report your prediction accuracy results along with a predictive modelling appropriate short discussion.
- Calculate the predictive importance of your predictors using the methods of permutation importance using the randomForest package in R. Discuss the results shortly in predictive modelling terms.
- Conclude with a short Remarks section where you summarise what you have done in your project and talk about any limitations in the way in which you have conducted your study and any future ideas for further development of your project.

Task 3 – Mark 10%

Aside from the random forest, do the following alternative model exploration.

- Train and test a simpler model (e.g. linear regression or decision tree) and compare its interpretability and accuracy to the Random Forest? Discuss the trade-off between model interpretability and performance in your financial problem setting.
- Train and test a neural network on your dataset and compare its performance to the random forest model in terms of prediction accuracy. Discuss the results.

Task 4 – Mark: 20%

Prepare and perform a 15min (incl. Q&A) long presentation of your research project. The presentation can be made by a group member of your choice (or more than one if you decide so), but in any case, **all group members should be in class** to answer questions. If this is not possible for serious reasons, please consult the lecturer well in advance.

Quality and clarity of coding – Mark: 10%

